

ENGINEERING DRAWING

I B.TECH - II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME02	ESC	L	T	P	C	CIE	SEE	Total
		1	-	3	2.5	40	60	100
COURSE OBJECTIVES The course will enable the students to: 1. Aimed to develop basic graphic skills to enable them to use for the preparation of engineering drawings, their readings and interpretations 2. To prepare the students to use technical skills and modern engineering tools necessary for the engineering practices 3. To exposure to a CAD package								
COURSE OUTCOMES At the end of the course the student should be able to: 1. Understand various commands and create drawing in AutoCAD. 2. Construct various engineering curves and know their importance. 3. Prepare orthographic projections of objects by visualizing them in different positions. 4. Solve the problem of projections of planes and solids in different positions. 5. Construct the isometric view into orthographic views and vice versa.								
UNIT - I	INTRODUCTION TO ENGINEERING DRAWING					CLASSES: 10		
Introduction to Engineering Drawing: Principles and their significance. Introduction to Computer Aided Drafting: Initial Setup Commands, Draw Commands, modify commands, 2D Drawings-Simple Exercises.								
UNIT - II	ENGINEERING CURVES					CLASSES: 10		
Engineering Curves: Ellipse, Parabola, and Hyperbola (General Method only), Involute.								
UNIT - III	ORTHOGRAPHIC PROJECTIONS OF POINTS AND LINES					CLASSES: 12		
Principles of Orthographic Projections –Projections of points. Projection of lines inclined to both planes (First angle projection only).								
UNIT - IV	PROJECTIONS OF PLANES AND SOLIDS					CLASSES: 10		
Projections of Planes: Projections of regular planes inclined to one plane. Projection of Solids: Solids inclined to one plane (Regular solids).								
UNIT - V	ISOMETRIC VIEW AND ORTHOGRAPHIC VIEWS - CONVERSION					CLASSES: 10		
Isometric view: Drawing Isometric circles, Dimensioning Isometric Objects. Conversion of Isometric view to Orthographic views and Orthographic to isometric view.								

TEXT BOOKS

1. Bhatt N.D., Panchal V.M. & Ingle P.R., Engineering Drawing, Charotar Publishing House.
2. Agrawal B. & Agrawal C. M., Engineering Graphics, TMH Publication.

REFERENCE BOOKS

1. D.M. Kulkarni, A.P.Rastogi, A.K. Sarka "Engineering Graphics with AutoCAD" PHI publications.
2. Narayana, K.L. & P Kannaiah, Text book on Engineering Drawing, Scitech Publishers.
3. Shah, M.B. & Rana B.C., Engineering Drawing and Computer Graphics, Pearson Education.